

Final Report Expansion Gap Review

Review Bot

May 10, 2026

1 1. Executive Summary

The final report has successfully transitioned from a rough draft to a structured document with proper boundaries around its "research prototype" nature. However, compared to proposal2 and the progress report, it lacks depth in literature references, specific architectural rationale, and concrete implementation details.

2 2. Current Report Status

The report correctly adopts safe wording (e.g., "modular ZK-rollup benchmarking prototype"). It contains placeholder sections for results and acknowledges missing data. However, the bibliography is sparser than previous proposals, and several strong sections from proposal2 are absent.

3 3. Proposal1-to-Final-Report Coverage

Proposal1 provides excellent background on the blockchain trilemma, L1/L2 taxonomy, and storage/read/write scalability. Much of this is compressed or missing in the current final report's background chapter.

4 4. Proposal2-to-Final-Report Coverage

Proposal2 clearly articulated the ZK-rollup motivation, research gaps, RQs, and methodology. The final report should integrate proposal2's deeper literature references and its rigorous definition of architecture.

5 5. Progress-Report-to-Final-Report Coverage

The progress report contains critical implementation details (e.g., specific sequencer, executor, and submitter code-level implementations) that have not been fully migrated to the final report's implementation chapters.

6 6. Code/Docs-to-Final-Report Coverage

The final report still lacks explicit mapping to the codebase's command-line operations, exact configuration management strategies, and limitations documented in the source code (such as the `prev_state_root` logic and RISC0 ELF parse errors).

7 7. Reference Coverage Analysis

- **Proposal1 Bibliography:** Estimated 15-20 references.
- **Proposal2 Bibliography:** Estimated 20-25 references.
- **Progress Report Bibliography:** Estimated 10-15 references.
- **Final Report Bibliography:** Currently 12 references.

The final report is missing numerous relevant citations present in proposal2.

8 8. Missing References to Add

Several critical references related to sequencer decentralization, DA layers, and specific ZK proof systems (e.g., PLONK, Halo2) from proposal2 are absent from the final `references.bib` and must be migrated.

9 9. Missing Sections to Add

- In-depth zkEVM complexity spectrum. - Expanded Optimistic vs. ZK Rollup comparison table. - Mathematical/state-transition model checks in the implementation.

10 10. Sections That Are Shorter Than Proposal2

- Literature Review (specifically the taxonomy of related work). - Research Methodology (justification of chosen parameters).

11 11. Risky Claims and Safer Wording

The report has mostly corrected risky claims, but must maintain strict adherence to "configuration-level comparison, not production performance claim".

12 12. Results and Metrics Gap

Results are correctly marked as "pending completed run," avoiding hallucinated data.

13 13. Required Tables/Figures

See Matrix D for required table definitions.

14. Chapter-by-Chapter Fix Plan

Future edits should map proposal2 literature to Chapter 3, progress report implementation details to Chapter 7, and explicitly cite missing references across Chapters 1-4.

15. Priority TODO List

1. Migrate missing references from proposal2 to `references.bib`.
2. Expand Background chapter with proposal1 trilemma data.
3. Expand Implementation chapter with progress report specifics.

16. Exact File Mapping for Next Editing Task

See Matrix E.

A. Source Coverage Matrix

Source	Key useful content	Present?	Missing detail	Target	Priority
Proposal1	Trilemma taxonomy	Partial	Storage scalability	Background	Med
Proposal2	ZK motivation, Lit Review	Partial	Deep reference citations	Lit Review	High
Progress Rpt	Implementation specifics	Partial	Sequencer/Executor deep-dive	Implementation	High

B. Reference Coverage Matrix

Reference	In Prop2/Prog?	In Final?	Cited?	Why needed	Action
Various ZK hardware refs	Yes (Prop2)	No	No	Contextualizes prototype	Migrate
DA off-chain protocols	Yes (Prop2)	Partial	Yes	Explains blob vs off-chain	Add

C. Proposal2 Stronger-than-Final Matrix

Prop2 Section	Why Stronger	Final Weakness	What to add	Target File
Lit Review	More comprehensive citations	Too brief	Migrate taxonomy	lit_review.tex
Architecture	Deep logical separation	Generic	Add interaction detail	architecture.tex

D. Progress Implementation Gap Matrix

Component	Progress Detail	In Final?	Missing	Target File
Sequencer	Mempool logic	Partial	Code-level flow	sequencer.tex
Executor	Merkle checks	No	Mathematical STF checks	executor.tex

E. Risky Claim Matrix

Claim/Location	Problem	Evidence	Safer Wording
Performance Results	Implies mainnet	Code	"Configuration-level comparison"

F. Results Availability Matrix

Metric	Claimed?	Found?	Safe Treatment	Action
Throughput/Latency/Cost	Yes	No	"Pending completed run"	Execute benchmarks